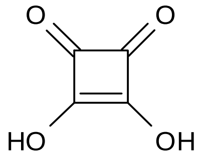
For Examiner's Use

1 Planning (P)

Squaric acid, $C_4H_2O_4$, also called quadratic acid, is a dibasic organic acid that has four carbon atoms bonded together to approximately form a square.



When an aqueous solution of squaric acid is added to a solution of sodium hydroxide, the temperature of the resultant solution increases, as the neutralisation reaction is exothermic.

A student was given a solution, **FA1**, containing squaric acid. She was asked to determine the enthalpy change of neutralisation when a solution, **FA2**, containing 2.0 mol dm^{-3} of sodium hydroxide was reacted with squaric acid of approximately the same concentration.

She decided to carry out a series of experiments, where different volumes of **FA1** and **FA2**, which together gave a total volume of 100 cm^3, were reacted. The temperature change, ΔT, determined from each experiment would be plotted against the volume of **FA1** used.

After carrying out a number of trial experiments, she found that for a typical experiment, the temperature of the resultant solution rose from the ambient room temperature of 29 °C to 35 °C. She correctly decided that the thermometer with divisions of 1 °C that she was using was inappropriate, and she asked her tutor for a more appropriate one.

(a) Write a balanced equation to represent the enthalpy change of neutralisation between squaric acid and sodium hydroxide.

[1]

(b) Explain briefly why the thermometer with 1 °C divisions that the student had used for the trial experiments was not appropriate.

[1]

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(c) Write a plan for **<u>one</u>** of the experiments of the series that the student will carry out.

You may assume that you are provided with the following.

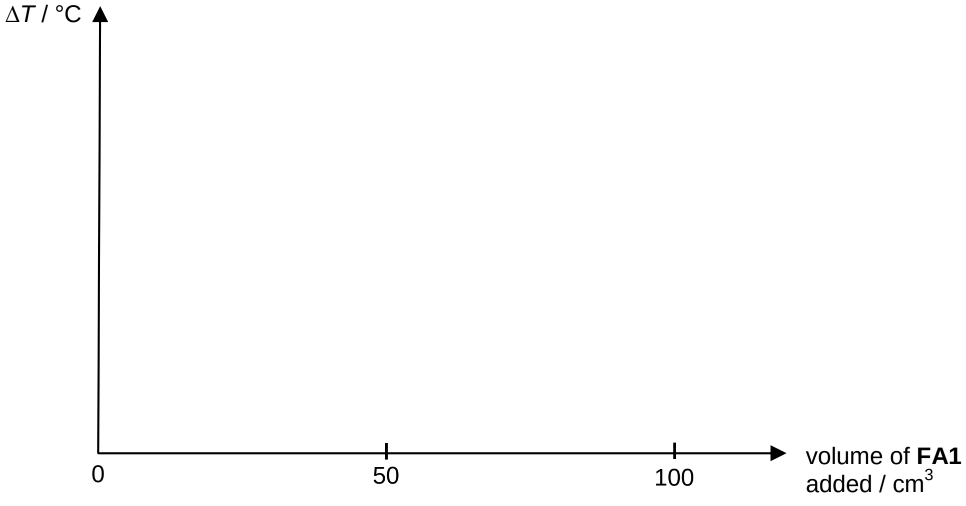
- o **FA1** and **FA2** solutions having the concentrations given above
- o 200 cm^3 styrofoam cup with lid
- o thermometer with divisions of 0.2 °C
- o the apparatus normally found in a school or college laboratory

Your plan should contain the following.

- o essential details, including appropriate quantities of the solutions used, and procedures to minimise experimental errors
- o a description of how the initial and final temperatures of the reaction mixture, and the temperature rise are determined

[4]

(d) Sketch on the axes below, the shape of the graph that will be obtained by plotting the values from the series of experiments, ignoring the effects of heat loss.



[1]

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(e) On the graph in part **(d)**, the coordinates of a **single** point on the graph will enable the student to calculate the concentration of squaric acid, and the enthalpy change of neutralisation of the reaction.
Indicate this point on the graph by labelling the values on the vertical and horizontal axes as **T** °C and **V** cm^3 respectively.

[1]

(f) Outline how you would use the results, **T** °C and **V** cm^3, that was obtained from part **(e)** to determine:

(i) the concentration of squaric acid in **FA1** in mol dm^{-3}.

(ii) the enthalpy change of neutralisation for the reaction in kJ mol^{-1}.

You may assume that 4.2 J of heat energy is required to raise the temperature of 1 cm^3 of the reaction mixture by 1 °C.

[3]

(g) Identify **one** potential safety hazard in this experiment and state how you would minimise this risk.

[1]

[Total: 12 marks]

For Examiner's Use

2 (a) Elements **J**, **K**, and **L** are consecutive elements in period 3 of the *Periodic Table*.

The first eight ionisation energies, *I.E.*, of element **K** are as follows.

I.E. / kJ mol^{-1}	1011	1907	2914	4963	6273	21267	25431	29872

(i) With reference to the data given above, deduce the Group of the *Periodic Table* to which **K** is likely to belong to.

(ii) Predict, giving your reasoning, whether the first ionisation energy of element **L** will be higher or lower than that of **K**.

(iii) When dissolved in water, the chlorides of elements **J** and **K** produce solutions that turn blue litmus red. Write balanced equations to represent the reactions of these chlorides with water.

chloride of **J**:

chloride of **K**:

[6]

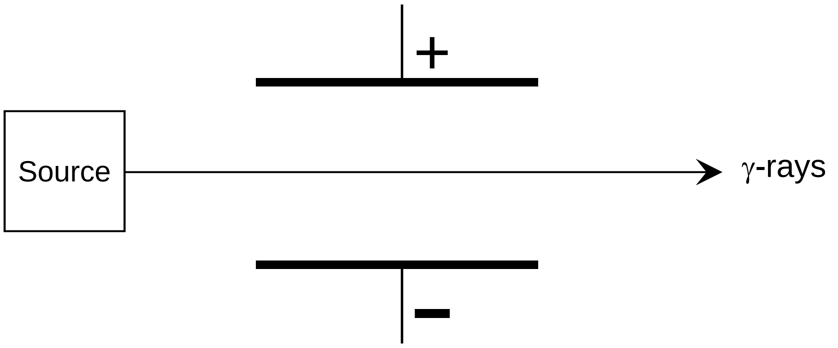
[Turn over

For Examiner's Use

(b) Chlorine occurs naturally with two stable isotopes, ^{35}Cl and ^{37}Cl, and one radioactive isotope, ^{36}Cl. Radioactive isotopes are unstable and they typically lose energy by emitting radiation through the process of radioactive decay.

Ernest Rutherford and later Paul Villard discovered the three types of radiation: α-particles (nucleus of helium atom); β-particles (electron); and γ-rays (electromagnetic wave).

(i) A stream of α-particles, β-particles and γ-rays is subjected to an electric field as shown below. The path taken by the γ-rays has already been drawn. Sketch on the diagram, how the α-particles and β-particles are affected by the electric field, given that they are travelling at the same speed. Label your answers clearly.



(ii) State the number of protons and neutrons in a nucleus of a chlorine-36 atom.

protons: ________________ neutrons: ________________

(iii) Chlorine-36 undergoes decay to form argon-36 as shown in the decay scheme:

$$^{36}Cl \rightarrow {}^{36}Ar + \textit{radiation Y}$$

Identify *radiation Y* and briefly explain how you arrive at your answer.

radiation Y: ________________________

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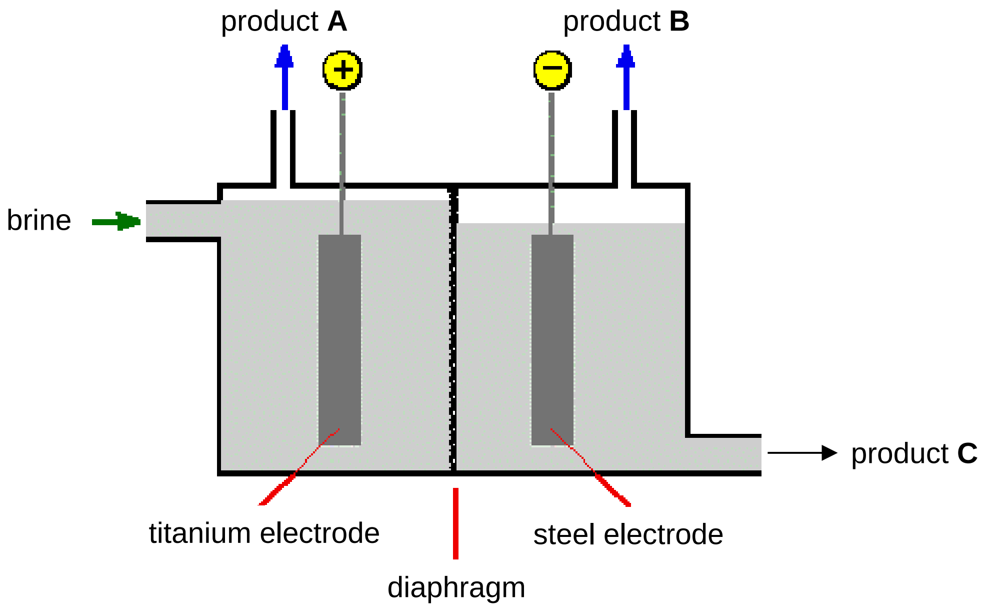
(iv) The relative amount of chlorine-36 and argon-36 in ancient samples of ground water is commonly used to determine the age of the water. This process is known as chlorine-36 dating.

In a sample of ice that was excavated from deep within a massive ice-berg near the Arctic Circle, it was found that the ratio of the amount of chlorine-36 to argon-36 is 1 : 7. Given that the half-life of the radioactive decay of chlorine-36 is 300,000 years, and that the decay is a first order reaction, deduce the age of the sample of ice.

[7]

For Examiner's Use

(c) Chlorine is manufactured commercially by the electrolysis of brine (concentrated sodium chloride) using the diaphragm cell.



(i) Give half equations for the two electrode processes that occur.

titanium electrode: ____

steel electrode: ____

(ii) Calculate the current that needs to be passed through the diaphragm cell in order to produce 1.00 tonne of chlorine gas per day.
(1 tonne = 1000 kg)

For Examiner's Use

The diaphragm is made of a porous mixture of asbestos and polymers, designed to prevent the products of the electrolytic cell from mixing with each other.

(iii) Write a balanced equation for the reaction that will occur if **A** is allowed to mix with **B**. Describe a potential safety hazard of this reaction.

equation: ____

potential safety hazard: ____________________________________

____

(iv) Write a balanced equation for the reaction that will occur if product **A** is allowed to mix with product **C** at room temperature.

____

[7]

(d) Chloride ions and bromide ions differ in their reactions with concentrated sulfuric acid.

Describe what you would see if concentrated sulfuric acid is added to separate samples of the solids KC*l* and KBr. Suggest an explanation for the differences in reaction, and write equations illustrating the types of reaction undergone.

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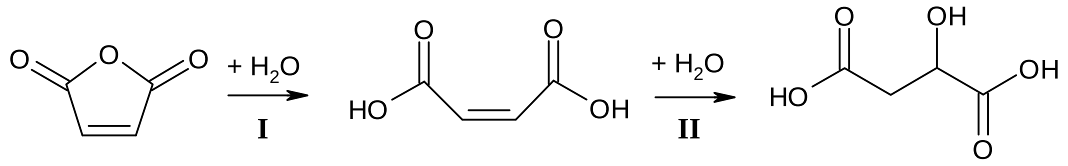
[3]

[Total: 23 marks]

For Examiner's Use

3 Malic acid, $HO_2CCH_2CH(OH)CO_2H$, was first isolated from apple juice by Carl Wilhelm Scheele in 1785, and is now commonly added to food products to produce a "tartness" in various food products like candy and beverages.

(a) Malic acid is produced industrially by reacting malic anhydride with water in a two-step reaction scheme as shown.



malic anhydride — **X** — malic acid

(i) The reactions of water in the two steps are markedly different. State the type of reaction in both steps.

Step **I**: ____

Step **II**: ____

(ii) Suggest reagents and conditions that would enable step **II** to be carried out.

____

(iii) The intermediate product, **X**, obtained from step **I** exist only in the cis-isomeric form. Suggest why this is so.

____

____

____

(iv) Describe a simple chemical test to distinguish between **X** and malic acid, stating the expected observations of each compound.

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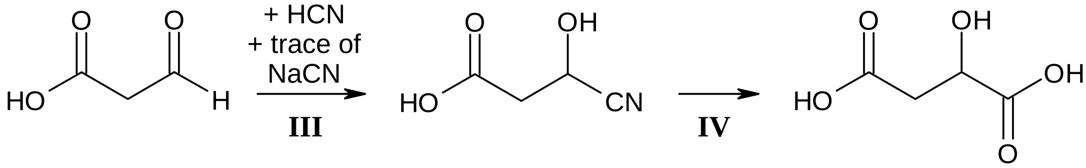
[6]

For Examiner's Use

(b) Malic acid can be synthesised in the laboratory by a different two-step reaction scheme from 3-oxopropanoic acid.



3-oxopropanoic acid

malic acid

The results of an investigation into the kinetics of step **III** of the reaction scheme are given below:

experiment number	initial concentration / mol dm^{-3}			relative initial rate
	[HO_2CCH_2CHO]	[HCN]	[NaCN]	
1	0.400	0.100	0.005	0.833
2	0.300	0.100	0.005	0.625
3	0.400	0.300	0.005	0.833
4	0.300	0.400	0.008	1.000

(i) Use these data to deduce the order of reaction with respect to each of the three reagents, showing how you arrive at your answers.
Hence write a rate equation for the reaction.

For Examiner's Use

(ii) Based on your results from **(b)(i)**, deduce the value of the relative initial rate of the reaction if the volume of the reaction mixture in **experiment 4** was doubled by the addition of water.

(iii) Describe a mechanism that is consistent with the rate equation that you have given in **(b)(i)**, and indicate the slowest step of the mechanism. Explain your reasoning.

(iv) State the difference between the malic acid synthesized using this reaction scheme and that obtained from apple juice. By considering the mechanism you have proposed in part **(iii)** and the shape of HO_2CCH_2CHO, account for this difference.

[11]

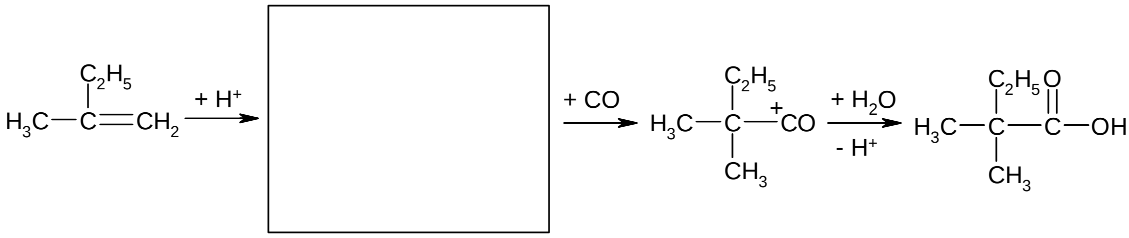
[Total: 17 marks]

For Examiner's Use

4 (a) Alkenes are susceptible to attack by electrophiles due to the presence of the electron-rich carbon-carbon double bond.

In the Koch reaction, alkenes react with carbon monoxide and water to form certain secondary and tertiary carboxylic acids.

The mechanism steps below shows 2-methylbut-1-ene undergoing the Koch reaction.



2-methylbut-1-ene | intermediate **P** | intermediate **Q** | product **R**

(i) Draw, in the space provided above, the structure of intermediate **P**.

(ii) Give the IUPAC name of the product **R**.

____

(iii) Deduce the number of electrons around the carbon atom that is bonded to the oxygen atom in intermediate **Q**.

number of electrons: ________________

(iv) Draw the displayed formula for the starting alkene that will form $CH_3CH_2CH(CH_3)CO_2H$ after undergoing the Koch reaction.

[4]

(b) A cycloalkane with the molecular formula C_5H_{10} gives only **two** mono-brominated products when reacted with bromine in the presence of ultraviolet light. Give the structural formulae of the two bromocycloalkane, and suggest the ratio in which the two products will be formed.

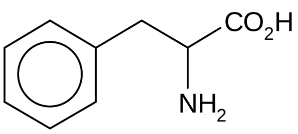
[3]

[Total: 7 marks]

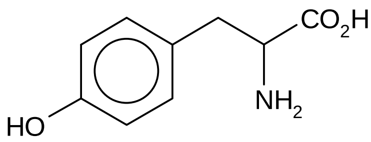
For Examiner's Use

5 Enzymes are large protein molecules that adopt a highly specific three-dimensional structure. They are highly selective catalysts that speed up many metabolic reactions, from the digestion of food to the synthesis of DNA.

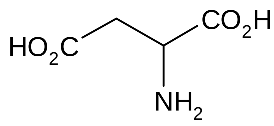
(a) The structural formulae of the three most common amino acids present in the polypeptide chain in the insulin-degrading enzyme are shown below.



phenylalanine (phe)



tyrosine (tyr)



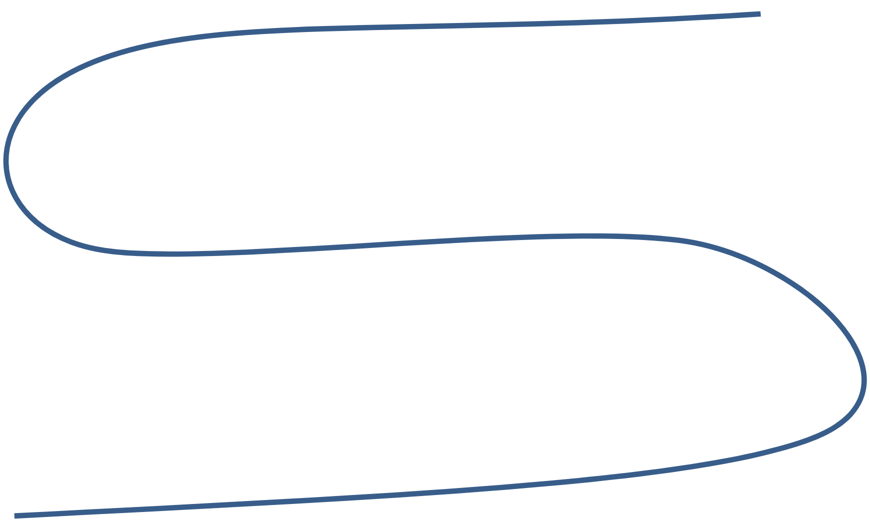
aspartic acid (asp)

(i) Draw the structural formula of the tripeptide phe-tyr-asp.

(ii) On the outline of a polypeptide chain below, illustrate the type of side chain interactions that will occur between the following pairs of amino acid residues.

- a tyrosine and a aspartic acid residue
- a phenylalanine and a tyrosine residue

Name the type of side-chain interactions that you have illustrated.



[3]

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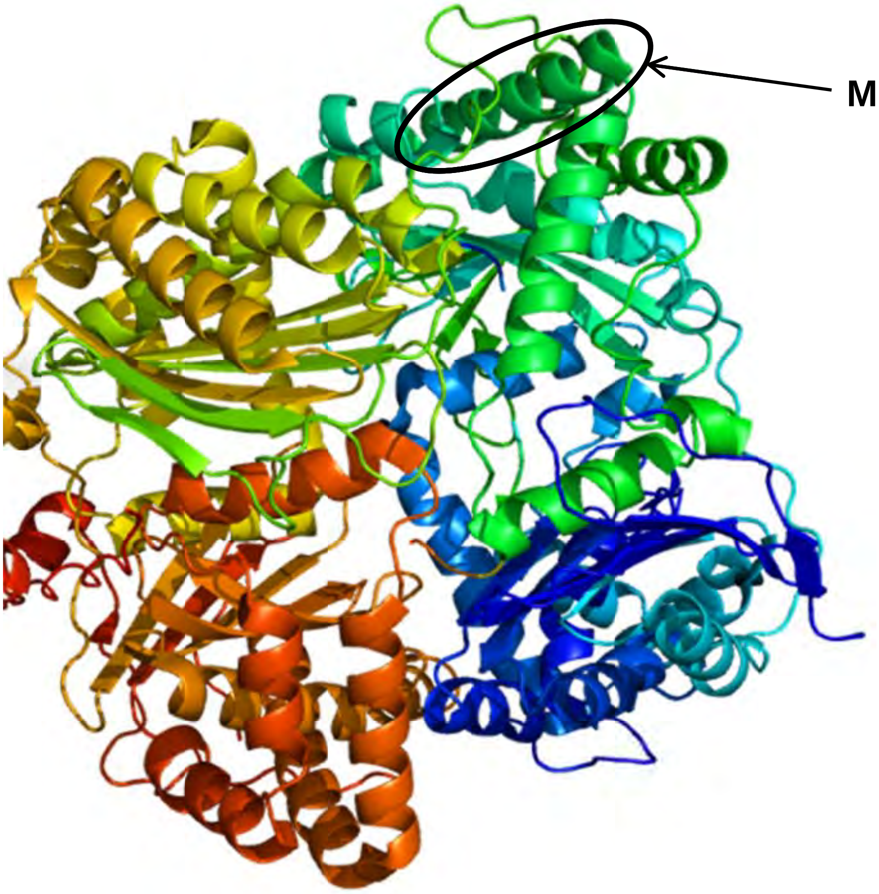
(b) The insulin-degrading enzyme is irregularly folded consisting of a large portion of α-helices, and regions of β-pleated sheets. A portion of the three-dimensional structure of the enzyme is represented below.

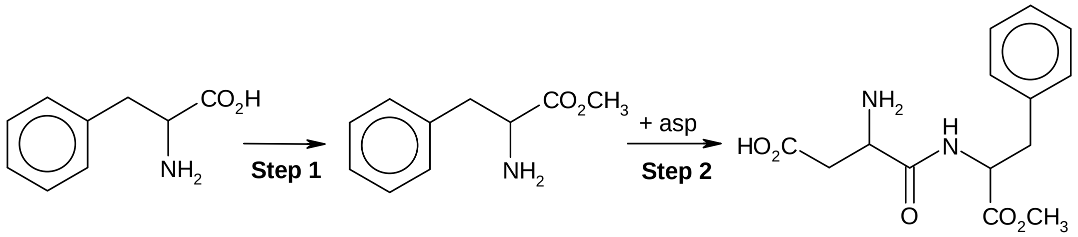


Describe the structure represented by the portion **M**, including how the structure is stabilised. You may wish to include a diagram in your description.

[2]

For Examiner's Use

(c) Aspartame, an artificial sweetener, can be synthesised from phenylalanine and aspartic acid as shown in the reaction scheme below.



phenylalanine aspartame

(i) Give the reagents and conditions for **Step 1**.

____

(ii) Suggest a reason why aspartame cannot be synthesised by reacting phenylalanine, aspartic acid, and the reagents in **Step 1** together in a one-step synthesis.

____

____

____

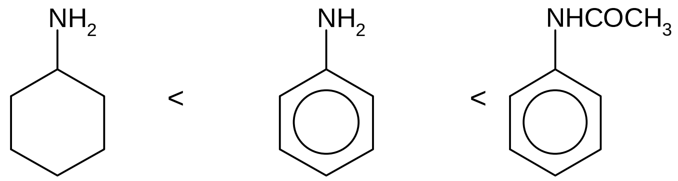
[2]

For Examiner's Use

(d) Explain each of the following observations as fully as you can.

(i) The melting point of glycine, H_2NCH_2COOH, (230 − 235 °C) is higher than that of 2-hydroxyethanamide, $HOCH_2CONH_2$ (102 − 104 °C).

(ii) The pK_b of the following compounds:



(iii) Nitration of methylbenzene, $C_6H_5CH_3$, produces the 1,2- and 1,4-substituted products as the major products, whereas nitration of *tert*-butylbenzene, $C_6H_5C(CH_3)_3$ gives predominantly the 1,4-substituted product.

[7]

[Total: 14 marks]

~ END OF PAPER ~